SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1702**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3 **Duration : 1 Hour**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I SAI GOPAL REDDY . Y(192524083)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
SAI GOPAL REDDY

QUESTION 1

A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

D1 cannot work Night.

D2 must work before D3, where Morning < Afternoon < Night.

Only one doctor can be assigned to each shift.

D3 cannot work Morning.

Every doctor must be assigned exactly one shift.

Problem Understanding

This is a constraint satisfaction problem (CSP). The variables are the three doctors, the domain of each variable is the set of shifts, and the constraints restrict permissible assignments. The final assignment must be complete, must use every shift exactly once, and must satisfy both individual and ordering constraints.

Variable	Initial domain	Relevant constraint
D1	{Morning, Afternoon, Night}	D1 â‰ Night, so D1 âˆˆ {Morning, Afternoon}
D2	{Morning, Afternoon, Night}	D2 must occur earlier than D3
D3	{Morning, Afternoon, Night}	D3 â‰ Morning, so D3 âˆˆ {Afternoon, Night}

Method Selection

Backtracking Search is appropriate because it assigns one variable at a time, checks constraints immediately, and returns to the most recent decision when a partial assignment cannot be extended. This avoids testing all 3! = 6 schedules blindly and gives a complete systematic search for this finite CSP.

Backtracking Search with intermediate steps

Use the variable order D2 â†’ D3 â†’ D1. The order is useful because the precedence relation directly involves D2 and D3.

Step	Partial assignment	Constraint checks	Decision
0	{ }	No assignment yet.	Start search.
1	D2 = Morning	D2 is assigned a valid shift.	Continue.
2	D2 = Morning, D3 = Afternoon	D3 is not Morning; Morning < Afternoon, so D2 < D3 is satisfied; shifts are distinct.	Continue to D1.

3 D1 = Night Distinct shift check passes, but D1 = Night violates D1 â‰ Night. Failure. Backtrack D1.

4 D2 = Morning, D3 = Afternoon D1 has remaining choices Morning or Afternoon, but both are already occupied. No legal value for D1. Backtrack D3.

5 D2 = Morning, D3 = Night D3 is not Morning; Morning < Night, so D2 < D3 is satisfied; shifts are distinct. Continue to D1.

6 D1 = Afternoon, D2 = Morning, D3 = Night D1 â‰ Night; all three shifts are distinct; D2 is before D3; D3 â‰ Morning. Success. Stop.

Final valid schedule

Shift	Assigned doctor	Verification
Morning	D2	D2 is before D3.
Afternoon	D1	D1 is not assigned Night.
Night	D3	D3 is not assigned Morning.

Final answer: Morning â†’ D2, Afternoon â†’ D1, Night â†’ D3. All doctors are assigned exactly once, every shift contains one doctor, D1 avoids Night, D3 avoids Morning, and D2 works before D3.

Level 4 (Exemplary) interpretation

The method demonstrates complete problem understanding, justified selection of backtracking, a logical sequence of assignments, explicit constraint checks, accurate elimination of the failed branch, and a final schedule whose feasibility is verified against every requirement.

QUESTION 2

A robot operates in a 5Ã—5 grid and must move from S to G. Movement is allowed Up, Down, Left and Right; every move costs 1; obstacles cannot be crossed; and Manhattan Distance is used as the heuristic. Use BFS to show node expansion, frontier/queue updates, and determine the optimal path and cost.

Representative grid assumption

The question does not provide obstacle coordinates, so the following representative 5Ã—5 grid is used. This assumption is stated explicitly so that the BFS expansion order and path can be reproduced exactly. The symbols are: S = Start, G = Goal, # = obstacle, and . = free cell.

Row \ Column	1	2	3	4	5
1	S	.	#	.	.
2	.	.	#	.	#
3	#	.	.	.	.

4 . # # . .

5 . . . # G

Problem formulation and heuristic

State: a cell represented as (row, column).

Initial state: S = (1,1).

Goal state: G = (5,5).

Actions: Up, Right, Down, Left, provided the destination is within the grid, not an obstacle, and not already visited.

Step cost: 1, so path cost equals number of moves.

Manhattan heuristic: h(r,c) = |r â^’ 5| + |c â^’ 5|.

BFS uses a FIFO queue and is optimal here because every move has identical cost 1. The Manhattan value is recorded for informed analysis and tie interpretation; BFS does not replace FIFO ordering with h(n), because doing so would be Greedy Best-First Search rather than BFS.

Node expansion and queue updates

Neighbor order used for reproducibility: Up, Right, Down, Left. A node is marked visited when it is enqueued, preventing duplicate insertion.

Step	Expanded node	h(n)	New legal nodes	FIFO queue after expansion
0	â€”	8	Start (1,1)	[(1,1)]
1	(1,1)	8	(1,2), (2,1)	[(1,2), (2,1)]
2	(1,2)	7	(2,2); right is obstacle	[(2,1), (2,2)]
3	(2,1)	7	None; (3,1) is obstacle and (2,2) visited	[(2,2)]
4	(2,2)	6	(3,2); (2,3) is obstacle	[(3,2)]
5	(3,2)	5	(3,3); (4,2) is obstacle	[(3,3)]
6	(3,3)	4	(3,4); (2,3), (4,3) are obstacles	[(3,4)]
7	(3,4)	3	(2,4), (3,5), (4,4)	[(2,4), (3,5), (4,4)]
8	(2,4)	4	(1,4); (2,5) is obstacle	[(3,5), (4,4), (1,4)]
9	(3,5)	2	(4,5); (2,5) is obstacle	[(4,4), (1,4), (4,5)]
10	(4,4)	2	None; (5,4) is obstacle and (4,5) visited	[(1,4), (4,5)]
11	(1,4)	5	(1,5)	[(4,5), (1,5)]
12	(4,5)	1	(5,5) = G	[(1,5), (5,5)]

13 (1,5) 4 None [(5,5)]

14 (5,5) = G 0 Goal reached; stop. []

Path reconstruction and cost

Using parent pointers from the goal gives:

(1,1) â†’ (1,2) â†’ (2,2) â†’ (3,2) â†’ (3,3) â†’ (3,4) â†’ (3,5) â†’ (4,5) â†’ (5,5)There are eight moves, each costing 1:

Total cost = 1+1+1+1+1+1+1+1 = 8The Manhattan distance from S to G is |1âˆ’5|+|1âˆ’5| = 8. Since the path avoids all obstacles and achieves this lower bound, it is optimal. BFS is therefore both complete and optimal for this assumed unit-cost grid.

Level 4 (Exemplary) interpretation

The answer states the missing-grid assumption, formally defines the search problem, explains why BFS is selected, records every queue update, uses Manhattan Distance correctly, reconstructs the path, verifies obstacle avoidance, and proves optimality by matching the Manhattan lower bound.

QUESTION 3

An autonomous rescue robot operates in a disaster-affected building to locate and rescue a survivor. The environment is partially observable, some paths are blocked, and decisions must be made with limited information. The robot starts at S and must reach a survivor at G.

(a) Constraint analysis and effect on decision-making

Constraint Effect on the robot's decision-making

Four-direction movement The action set is discrete: Up, Down, Left and Right. Diagonal shortcuts are not permitted, so the route must be composed of legal orthogonal moves.

Obstacles Blocked cells must be treated as impassable. The robot must maintain a map or belief about blocked cells and eliminate unsafe successors.

Limited sensing The robot cannot know the whole building initially. It must use adjacent-cell observations to update its internal map and make decisions online.

Movement cost 1 Each ordinary move contributes one unit to the total cost, so shorter safe routes are preferred.

Risk-zone surcharge 2 Entering a risky cell has total step cost 1 + 2 = 3. A longer safe route can therefore be better than a shorter risky route.

Minimum cost and safety The objective is not simply minimum distance; it is minimum accumulated cost subject to safe traversability.

Dynamic knowledge New observations can change the map. The robot must re-plan when it discovers an obstacle, risk zone or a newly opened passage.

(b) Solution approach

A suitable solution is a model-based online search agent using Uniform Cost Search (UCS) over the robot's current belief map. UCS is appropriate because costs are non-negative and risk zones have different costs.

Initialize the robot's knowledge map with S, G if known, unknown cells, and observed obstacles/risk cells.

Sense adjacent cells and update the internal map.

Generate legal successors: in-bounds, unblocked and safe-enough cells.

Assign edge cost 1 to an ordinary cell and 3 to a risky cell.

Run UCS using cumulative cost g(n), always expanding the known frontier state with the lowest g(n).

When a route is selected, execute only the next safe action rather than blindly committing to the entire route.

Sense again, update the map, and re-plan if the new percept conflicts with the previous plan.

Stop when G is reached and the survivor is safely located; report the accumulated cost and any risk encountered.

PROCEDURE OnlineRescue(S, G):

map â†□ initialiseBeliefMap()

current â†□ S

WHILE current â‰ G DO

percepts â†□ SenseAdjacentCells(current)

map â†□ UpdateMap(map, percepts)

frontier â†□ priority queue ordered by cumulative cost g

INSERT (current, cost=0) into frontier

parent â†□ empty map

bestCost[current] â†□ 0

WHILE frontier is not empty DO

n â†□ REMOVE-MIN(frontier)

IF n = G THEN

route â†□ Reconstruct(parent, G)

BREAK

FOR each legal neighbour m of n in map DO

IF m is blocked THEN CONTINUE

stepCost â†□ 1

IF m is risky THEN stepCost â†□ 3

newCost â†□ bestCost[n] + stepCost

IF m is not known OR newCost

(c) AI concepts demonstrated

AI concept Application in the rescue robot

Intelligent agent The robot perceives adjacent cells, maintains an internal model, selects actions, moves through the building, and acts to achieve the goal of reaching the survivor.

Rationality For the available information, the robot chooses the action that minimizes expected accumulated cost while respecting safety. It does not choose a blocked cell or a locally short route that produces greater risk cost.

Partial observability The true building state is not fully visible. The robot reasons using a belief/map that becomes more accurate through sensing.

Online search Planning and execution are interleaved. The robot observes, plans a safe next action, executes, and replans when information changes.

Utility/cost-sensitive reasoning Movement and risk are combined into a numerical objective, allowing a safe longer route to be preferred over a dangerous shorter route.

Environment characterization

The environment is partially observable, deterministic for known actions, sequential, dynamic, discrete, unknown initially, and single-agent from the robot's planning perspective. It is partially observable because only adjacent cells are sensed; sequential because each move changes the robot's future position; dynamic because hazards or blocked passages may change; discrete because rooms/cells and actions are discrete; and unknown initially because the complete map is unavailable.

(d) Justification of the chosen approach

UCS is more suitable than ordinary BFS because the surcharge makes step costs unequal: ordinary movement costs 1 whereas entering a risk zone costs 3. BFS minimizes the number of moves, not the total risk-adjusted cost. A purely offline plan is also unsuitable because the robot cannot see the entire building in advance. The model-based online UCS agent combines optimal cost search over known information with continuous sensing and replanning. With non-negative costs, UCS is complete and returns the least-cost known route; online execution preserves safety when new obstacles or risks are discovered.

Level 4 (Exemplary) interpretation: The response identifies every constraint, selects and justifies an appropriate cost-sensitive online strategy, provides implementable pseudocode, accurately applies intelligent-agent and rationality concepts, characterizes the environment, and connects the solution to safe engineering operation under uncertainty.

FINAL SUMMARY OF LEVEL 4 COVERAGE

All three answers explicitly demonstrate: clear problem understanding, justified method selection, systematic solution process, accurate calculations or constraint checks, and meaningful interpretation with engineering relevance.